

Practical - 2

Aim: Console based node.js applications

1: Write a Node.js program that accept five numbers from command-line and compute their sum, average, smallest and largest number.

Code:

```
const args = process.argv.slice(2);

if (args.length !== 5) {
  console.error('Error: Please provide exactly 5 numbers.');
```

console.log('Usage: node compute.js <num1> <num2> <num3> <num4> <num5>');

process.exit(1);

}

const numbers = args.map(Number);

if (numbers.some(isNaN)) {

console.error('Error: All inputs must be valid numbers.');

process.exit(1);

}

const sum = numbers.reduce((acc, curr) => acc + curr, 0);

const average = sum / numbers.length;

const smallest = Math.min(...numbers);

const largest = Math.max(...numbers);

console.log('--- Results ---');

console.log(`Numbers: \${numbers.join(', ')}`);

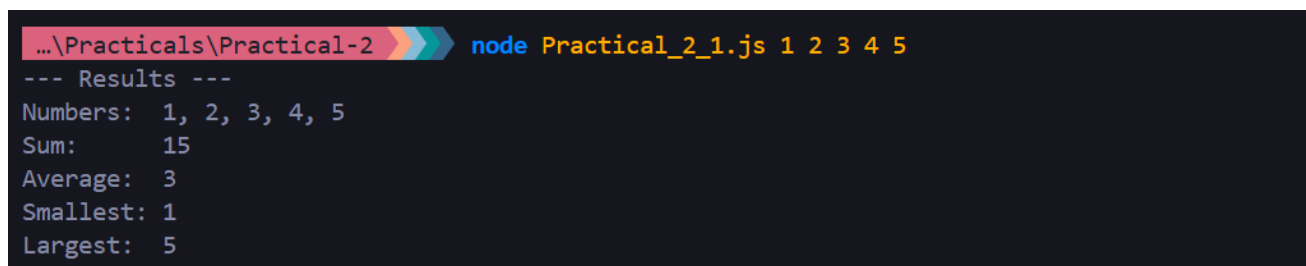
console.log(`Sum: \${sum}`);

console.log(`Average: \${average}`);

console.log(`Smallest: \${smallest}`);

console.log(`Largest: \${largest}`);

Output:



```
...\\Practicals\\Practical-2 > node Practical_2_1.js 1 2 3 4 5
--- Results ---
Numbers: 1, 2, 3, 4, 5
Sum: 15
Average: 3
Smallest: 1
Largest: 5
```

2: Create To-Do List App in Node.js using readline package that allows adding tasks one by one and viewing all tasks.

Code:

```

const readline = require('readline');

const rl = readline.createInterface({
  input: process.stdin,
  output: process.stdout,
});
const tasks = [];
function showMenu() {
  console.log('\n--- TO-DO LIST MENU ---');
  console.log('1. View All Tasks');
  console.log('2. Add a Task');
  console.log('3. Exit');

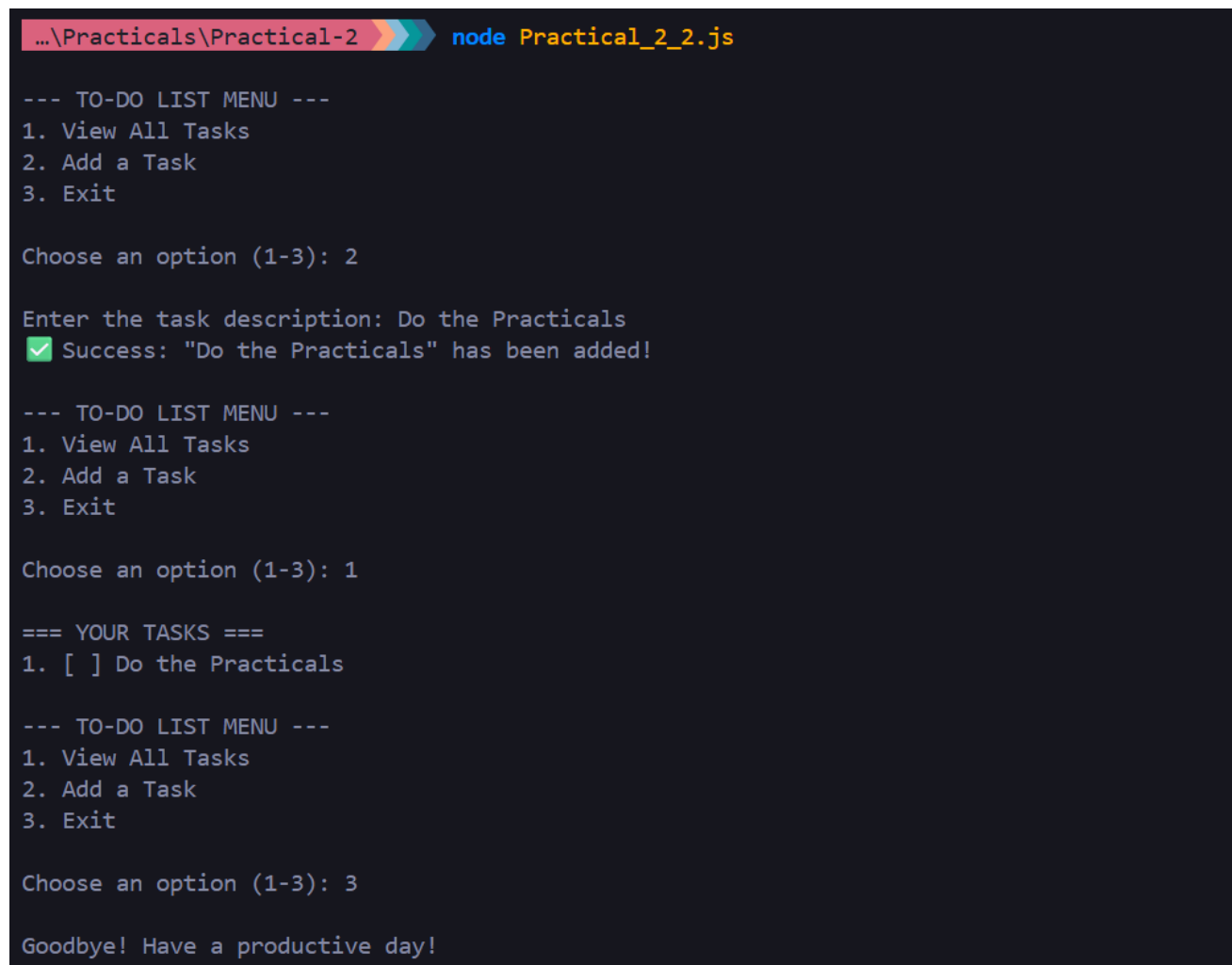
  rl.question('\nChoose an option (1-3): ', handleMenuInput);
}
function handleMenuInput(choice) {
  switch (choice.trim()) {
    case '1':
      viewTasks();
      showMenu();
      break;
    case '2':
      addTaskPrompt();
      break;
    case '3':
      console.log('\nGoodbye! Have a productive day!');
      rl.close();
      break;
    default:
      console.log('\n✗ Invalid choice! Please enter 1, 2, or 3. ');
      showMenu();
      break;
  }
}
function viewTasks() {
  console.log('\n=== YOUR TASKS ===');
  if (tasks.length === 0) {
    console.log('(No tasks added yet. Your list is clean!)');
  } else {
    tasks.forEach((task, index) => {
      console.log(`${index + 1}. [ ] ${task}`);
    });
  }
}
function addTaskPrompt() {
  rl.question('\nEnter the task description: ', (task) => {
    const cleanedTask = task.trim();
  });
}

```

```
    if (cleanedTask === "") {
        console.log('✗ Task cannot be empty!');
    } else {
        tasks.push(cleanedTask);
        console.log(`☑ Success: "${cleanedTask}" has been added!`);
    }

    showMenu(); // Return to the main menu
});
}

showMenu();
```

Output:

```
...\\Practicals\\Practical-2 > node Practical_2_2.js

--- TO-DO LIST MENU ---
1. View All Tasks
2. Add a Task
3. Exit

Choose an option (1-3): 2

Enter the task description: Do the Practicals
☑ Success: "Do the Practicals" has been added!

--- TO-DO LIST MENU ---
1. View All Tasks
2. Add a Task
3. Exit

Choose an option (1-3): 1

=== YOUR TASKS ===
1. [ ] Do the Practicals

--- TO-DO LIST MENU ---
1. View All Tasks
2. Add a Task
3. Exit

Choose an option (1-3): 3

Goodbye! Have a productive day!
```